

Gear Up Claw Master — Build Manual

GearUp · FTC Team #31242

This manual walks you through building the **Gear Up Claw Master** robot arm from start to finish. The arm has **4 moving joints**, all controlled wirelessly over WiFi. Build in order, Stage 1 → Stage 7, and don't skip the "Before You Start" section — it explains servo centering, which is the #1 thing people get wrong.

Joint	Code Channel	Servo	What it does
Base	CH 3	MG996R (360° continuous)	Spins the whole arm left and right
Shoulder (Lower)	CH 0 (<i>inverted</i>)	MG995 (180°)	Raises and lowers the lower section
Elbow (Arm)	CH 1	MG995 (180°)	Extends the middle section in and out
Claw	CH 2	MG995 (180°)	Opens and closes the gripper

Why CH 0 is "inverted": In the code, when you slide the Shoulder slider UP, the servo actually turns DOWN. This is done on purpose so the arm moves in the direction that feels natural. You don't need to do anything special — just know that's expected behavior.

Parts List (from the kit order)

3D-Printed Parts

- Main base box (the big square frame) — ×1
- Big ring gear (mounts on the lazy susan) — ×1
- Small gear (mounts on the continuous servo) — ×1
- Small base (the wide open rectangular plate that sits on top of the ring gear) — ×1
- Joint bracket (used for BOTH shoulder and elbow — same interchangeable part) — ×2
- Forearm link (connects the elbow to the wrist) — ×1
- Claw holder / wrist plate — ×1
- Gripper fingers (one driven finger + one idle finger) — ×2

Electronics

- ELEGOO ESP-32 development board — ×1
- HiLetgo PCA9685 16-channel servo driver — ×1
- MG995 metal-gear servo (180°) — ×3 used (you have a 4-pack, so ×1 spare)
- MG996R metal-gear servo (360° continuous) — ×1 used (you have a 2-pack, so ×1 spare)

Power

- LampVPath 4×AA battery holder with leads — ×1
- Rechargeable AA 2800mAh batteries + charger — ×4 batteries
- Amazon Basics USB-C to USB-A cable (for programming the ESP32) — ×1

Mechanical Hardware

- Dailydanny 4" aluminum lazy susan turntable bearing — ×1
- METERXITY F683ZZ flanged bearings (3mm × 7mm × 3mm) — for the claw idle pivot
- uxcell M3×25 pan-head Phillips bolts — long bolts (claw idle pivot)
- uxcell M3×12 pan-head Phillips bolts — servo mounts, horn-to-part joints, bracket joints
- M3 nuts — pair with the M3 bolts
- Krysay #8×¾" PAN-head wood screws — mounting the small base, and fastening into printed parts
- FIXLINK #8×¾" FLAT-head wood screws — fastening into printed parts where a flush head is needed

Tools — Exactly Which Screwdrivers to Use

Every fastener in this kit is a **Phillips (cross-head)** drive. The trick is matching the Phillips NUMBER to the screw, or you'll strip the head. Use this exact mapping:

Fastener	Driver to use
Krysay #8×¾" pan-head wood screws	#2 Phillips (PH2)
FIXLINK #8×¾" flat-head wood screws	#2 Phillips (PH2)
uxcell M3×12 pan-head bolts	#1 Phillips (PH1)
uxcell M3×25 pan-head bolts	#1 Phillips (PH1)

Fastener	Driver to use
Servo mounting-tab screws (came bagged with each servo)	#1 Phillips (PH1)
Servo horn CENTER screw (the one that holds the horn on the shaft)	#1 Phillips (PH1)
Servo horn OUTER screws (the tiny ones through the horn's small holes)	#0 Phillips (PH0)

Important catch about your screwdriver set: Many small "precision/electronics" screwdriver sets only go up to PH1 or PH00 — they are great for the servo screws and the M3 bolts, but they are **too small for the #8 wood screws**. The #8 wood screws need a **full-size #2 Phillips (PH2)**. Before you start, check that your set either includes a PH2 bit/driver or grab a standard #2 Phillips separately. If a driver wobbles in the screw head, it's the wrong size — go up or down one number until it sits snug with no wiggle.

Also handy: small needle-nose pliers to hold M3 nuts while you tighten the bolts.

Before You Start — Servo Centering (READ THIS)

The code starts **all three MG995 servos at 90°** (the neutral middle position). If you bolt a horn on while the servo is sitting at the wrong angle, the joint will only be able to move in one direction. So before mounting any horn:

1. Wire up the ESP32 + PCA9685 + battery temporarily (see Stage 6) and flash the code, OR use a simple servo-centering sketch.
2. Plug the servo into its channel and let it snap to **90°**.
3. NOW press the horn on, aimed the way you want that joint to point when it's centered.

The **MG996R (Base)** is a continuous-rotation servo — it spins instead of holding an angle, so it does NOT get centered. Just press its small gear straight onto the shaft.

Other rules:

- **Never force a bolt into a tight 3D-printed hole.** Turn it in slowly to cut its own threads, back it out once, then drive it home. Forcing it cracks the plastic.
- **Leave slack in every servo wire** so the joint can move without yanking the cable.
- **Charge your AA batteries fully** before the first power-up.

Stage 1 — The Base & Rotation (CH 3)

This is the foundation. Everything above it spins as one unit.

1.1 — Mount the lazy susan to the main base box. Center the **4" lazy susan turntable** on the top face of the **main base box**, lining its bottom plate up with the base's mounting holes. Drive **Krysay #8×¾" pan-head wood screws** (PH2) down through the lazy susan's bottom plate into the base. Tighten in a cross/star pattern (one screw, then the one across from it) so it seats flat and doesn't rock.

1.2 — Attach the big ring gear to the lazy susan. Set the **big ring gear** (teeth facing out, flat mounting face down) on the lazy susan's rotating TOP plate. Line up the outer bolt holes and fasten with **M3×12 bolts + M3 nuts** (PH1). Hold each nut underneath with pliers while you turn the bolt. When done, the whole ring gear should spin freely on the lazy susan with no wobble.

1.3 — Mount the MG996R continuous servo to the base. Position the **MG996R** so its output shaft is at the right height for its small gear to line up with the ring gear's teeth. Drop the servo body into its pocket/mount on the base, then secure it through its two mounting tabs with **M3×12 bolts + nuts** (PH1), or the long self-tapping screws that came bagged with the servo. Make sure the wire exits toward the inside of the base.

1.4 — Install the small gear on the MG996R. Press the **small gear** straight down onto the MG996R's splined output shaft, then lock it on with the servo's center screw (PH1). Check the **mesh**: the small gear's teeth should engage the big ring gear's teeth with just a hair of play — not jammed tight (it'll grind and burn out the servo) and not loose (it'll skip teeth). If the mesh is off, loosen the MG996R mounts, nudge it closer or farther, and re-tighten.

Test it: Spin the small gear by hand. The entire ring gear should rotate smoothly all the way around. Fix the mesh now — it's much harder once the arm is built on top.

Stage 2 — Mount the Small Base (on top of the gear)

The **small base** is the wide, open rectangular plate that sits on top of the ring gear. It's the platform the whole arm tower bolts to, and it rotates with the gear.

2.1 — Set the small base on the ring gear. Place the **small base** flat and centered on top of the **big ring gear**, lining its mounting holes up with the holes in the center of the ring gear.

2.2 — Fasten it with pan-head wood screws. Drive **Krysay #8×¾" PAN-head wood screws** (PH2) down through the small base into the ring gear. Use the pan-head (rounded top) screws here, not the

flat-head ones — the pan head sits on top and clamps the plate down. Tighten evenly so the small base sits dead flat with no gap.

Test it: Slowly rotate the small base by hand. It should turn the ring gear on the lazy susan smoothly, carrying the whole platform around. This is your Base (CH 3) rotation working.

Stage 3 — The Shoulder Joint (CH 0)

The shoulder is the first **joint bracket**, standing up from the small base.

3.1 — Mount the shoulder bracket to the small base. Stand one **joint bracket** upright on the **small base** in its mounting position. Bolt it down through the base holes with **M3×12 bolts + nuts** (PH1). Everything from here up will rotate with the base, so make sure the bracket is square and solid.

3.2 — Seat the shoulder servo. Take an **MG995** and drop it into the bracket's rectangular servo pocket. Orient it so:

- the **output shaft (the splined wheel)** points toward where the next joint will pivot, and
- the **wire exits downward**, toward the base, so it can run down to the PCA9685.

The servo's two mounting tabs should rest flat against the bracket's faces.

3.3 — Bolt the servo in. Drive the screws through both mounting tabs into the bracket — use the **self-tapping screws bagged with the servo** (PH1) or **M3×12 bolts + nuts** (PH1). Snug them down evenly so the servo can't shift, but don't overtighten into the plastic.

3.4 — Center it, then add the horn. Power the servo and let it go to **90°** (see "Before You Start"). Press the horn that came with the servo onto the shaft, aimed so the next bracket will point straight up at 90°. Lock the horn with its **center screw** (PH1). Leave the next part off for now — you'll attach it in Stage 4.

Stage 4 — The Elbow Joint (CH 1)

The elbow uses the **same joint bracket** as the shoulder (they're interchangeable).

4.1 — Attach the elbow bracket to the shoulder servo's horn. Position the second **joint bracket** against the shoulder servo's horn so the shoulder servo will swing this bracket up and down. Fasten the bracket to the horn using the **tiny horn screws** (PH0) through the horn's outer holes. This is the connection that lets the shoulder lift the elbow.

4.2 — Seat and bolt the elbow servo. Drop another **MG995** into the elbow bracket's pocket, output shaft facing toward where the forearm will pivot, wire exiting down toward the base. Bolt it in through the mounting tabs (PH1), same as Stage 3.3.

4.3 — Center it, then add the horn. Send this servo to **90°**, press its horn on aimed for "straight" at 90°, and lock it with the **center screw (PH1)**.

Stage 5 — The Forearm Link

The **forearm link** bridges the elbow to the wrist/claw.

5.1 — Mount the forearm to the elbow servo's horn. Attach the **forearm link** onto the elbow servo's horn so the elbow servo swings the whole forearm. Secure it with the **tiny horn screws (PH0)** through the horn holes. Give it a gentle test swing by hand-rotating the servo horn slightly (don't force it) to confirm it pivots cleanly.

Stage 6 — The Wrist & Claw (CH 2)

This is the gripper. One finger is **driven directly by the claw servo**; the other is an **idle finger** that mirrors it through meshing gears, pivoting on a bearing.

6.1 — Mount the claw holder to the forearm. Attach the **claw holder / wrist plate** to the end of the **forearm link** with **M3×12 bolts + nuts (PH1)**.

6.2 — Seat the claw servo. Center the last **MG995** to 90°, then set it into the **big hole** in the claw holder so its output shaft pokes through to the finger side. Bolt the servo body to the claw holder through its mounting tabs (PH1). This servo drives one finger directly.

6.3 — Install the idle-finger bearing. Press one **F683ZZ flanged bearing (3×7×3mm)** into the small bearing seat on the bottom of the claw holder until the flange sits flush. Slide a **long M3×25 bolt (PH1)** through the bearing's 3mm bore. Because the bearing spins freely, this bolt becomes a smooth, low-friction pivot for the idle finger. Secure it with an **M3 nut** on the back — tighten just until snug, NOT so tight that it pinches the bearing and stops it spinning.

6.4 — Attach the driven gripper finger. Mount the **driven finger** onto the claw servo's horn (the star/round horn) so the servo turns it directly. Use the **tiny horn screws (PH0)**. Its gear teeth need to sit at the same height as the idle finger's gear teeth.

6.5 — Attach the idle gripper finger and mesh the gears. Mount the **idle finger** onto the **M3×25 pivot bolt** so it spins on the **F683ZZ bearing**. Rotate it so its gear teeth **mesh** with the driven finger's gear teeth. When meshed correctly, turning one finger turns the other the opposite way — that's what makes both fingers open and close together.

Test it: Gently move the driven finger by hand. Both fingers should open and close symmetrically. If only one moves, the gears aren't engaged — lift the idle finger off, rotate one tooth, and re-mesh until both move together.

Stage 7 — Electronics, Wiring & Power-On

All the electronics surface-mount on the **outside of the main base box**.

7.1 — Mount the boards. Fasten the **ELEGOO ESP32** and the **HiLetgo PCA9685** to the outside of the base box using double-sided foam tape or **#8×¾" wood screws (PH2)** into small standoffs. Mount the **4×AA battery holder** on the base too. Keep the ESP32's USB-C port reachable from the edge so you can plug in to reprogram later.

7.2 — Wire the ESP32 to the PCA9685 (I2C — the 2-wire control bus).

ESP32 pin	→	PCA9685 pin
3.3V	→	VCC
GND	→	GND
GPIO 21 (SDA)	→	SDA
GPIO 22 (SCL)	→	SCL

The PCA9685 lives at I2C address **0x40**, which is exactly what the code expects.

7.3 — Power the servos from the battery pack. Load the **4 charged 2800mAh AA batteries** into the **battery holder** and connect its leads to the PCA9685's **V+ and GND power terminals** (the screw/header block on the side, NOT the I2C pins). The servos draw their muscle power from here. **Do NOT power the servos from the ESP32's pins** — servos pull far too much current and will brown out or damage the board.

7.4 — Plug the servos into the correct channels. Each servo has a 3-wire plug (signal / V+ / GND). Match the colored stripe to the channel orientation on the PCA9685, and plug each servo into the channel the code expects:

PCA9685 Channel	Servo	Joint
CH 0	MG995	Shoulder (Lower) — <i>inverted in code</i>
CH 1	MG995	Elbow (Arm)
CH 2	MG995	Claw
CH 3	MG996R	Base (rotation)

7.5 — Flash the code. Plug the **USB-C cable** from your computer into the ESP32 and upload the Gear Up Claw Master sketch.

7.6 — Connect to the arm and control it.

1. After it boots, the ESP32 creates its own WiFi network named **GearUp-Arm**.
2. On your phone, join that network. The password is **12345678**.
3. Open a browser and go to <http://192.168.4.1>.
4. You'll see four sliders — **Base, Lower, Arm, Claw**. Drag a slider and the matching servo moves. Everything starts centered at 90°.

Troubleshooting

- **A servo buzzes/jitters but barely moves** → it's fighting its mechanical limit. Re-center it to 90° and remount the horn (Before You Start).
- **The base won't rotate** → check the small gear ↔ ring gear mesh (Stage 1.4) and confirm the MG996R is plugged into **CH 3**.
- **Only one claw finger moves** → the gripper gears aren't meshed (Stage 6.5).
- **The idle finger won't move at all / is stiff** → the M3×25 nut is overtightened and pinching the F683ZZ bearing (Stage 6.3). Back it off slightly.
- **A joint moves the wrong way** → for the Shoulder, that's the intended inverted behavior (CH 0). For other joints, the horn was mounted off-center; re-center at 90° and remount.
- **Servos are weak or the ESP32 keeps resetting** → battery pack isn't wired to the PCA9685 V+/GND, or the AAs need charging.
- **Can't find the "GearUp-Arm" WiFi** → the code didn't flash or the ESP32 isn't powered. Re-upload over USB-C and check power.
- **A screw head is stripping** → wrong Phillips size. #8 wood screws need PH2; M3 bolts need PH1; tiny horn screws need PH0.

Built by GearUp · FTC Team #31242